

Retinal Disease Classification with Explainable AI: Multi-Label Diabetic Retinopathy and Macular Edema Grading

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Abstract

Diabetic retinopathy (DR) and diabetic macular edema (DME) are leading causes of preventable blindness and are typically graded jointly during clinical screening, yet most machine learning approaches treat DR grading as an isolated single-label task. This work presents a shared-backbone, dual-head deep learning model that jointly predicts DR severity (5-class, grades 0–4) and DME risk (3-class, grades 0–2) from a single fundus image, using fundus-specific preprocessing and Grad-CAM explainability. Trained on the IDRiD Disease Grading dataset with 5-fold stratified cross-validation, the model achieves a combined quadratic weighted kappa of 0.795 ± 0.038 . Grad-CAM analysis surfaces two clinically relevant findings: a systematic tendency to under-grade severe cases, and substantially overlapping attention between the DR and DME prediction heads, which runs counter to the initial hypothesis that the two tasks would localize to anatomically distinct regions.

1 Motivation

Retinal screening programs grade DR and DME together because the two conditions share underlying vascular pathology and are both required for a referral decision. A model that predicts only DR, as most published portfolio-scale projects do, does not reflect this clinical workflow. Beyond the joint-prediction gap, black-box severity grading is difficult to trust in a clinical setting: a clinician cannot act on a probability alone without knowing which visual evidence produced it. This project addresses both gaps by (1) framing DR and DME grading as a joint multi-label task on a shared backbone, and (2) attaching per-task Grad-CAM explainability so that predictions can be visually audited rather than taken at face value.

2 Dataset

The IDRiD (Indian Diabetic Retinopathy Image Dataset) Disease Grading subset was used, consisting of 413 training and 103 test fundus images, each labeled with both a DR severity grade (0: no DR – 4: proliferative DR) and a DME risk grade (0: no risk – 2: severe risk). This dataset was chosen specifically because it provides both labels natively, avoiding the label-quality risk of merging separate single-label datasets. The dataset is small by deep learning standards and DR grade 1 / DME grade 1 are underrepresented, both of which materially shape the results discussed in Section 5.

3 Method

3.1 Preprocessing

Each fundus image was processed with a pipeline specific to retinal photography rather than generic natural-image preprocessing:

- **Circular crop** to remove the black background surrounding the fundus disc.
- **CLAHE** (Contrast Limited Adaptive Histogram Equalization) applied to the green channel specifically, since retinal lesions exhibit the highest contrast in that channel.
- **Ben Graham normalization** [1], which subtracts a locally blurred version of the image to enhance the visibility of small lesions such as microaneurysms and hemorrhages. This method won the original Kaggle Diabetic Retinopathy Detection competition.

3.2 Architecture

A ResNet-50 backbone, pretrained on ImageNet, is shared across two independent linear classification heads: a 5-class DR head and a 3-class DME head (Figure 1). Sharing the backbone reflects the clinical correlation between the two conditions and is more data-efficient than training two separate networks given the dataset’s small size. Total loss is a weighted sum of the two task losses:

$$\mathcal{L} = w_{DR} \cdot \mathcal{L}_{CE}^{DR} + w_{DME} \cdot \mathcal{L}_{CE}^{DME} \quad (1)$$

with $w_{DR} = 1.0$, $w_{DME} = 0.7$, and per-class weights applied within each cross-entropy term to counter class imbalance.

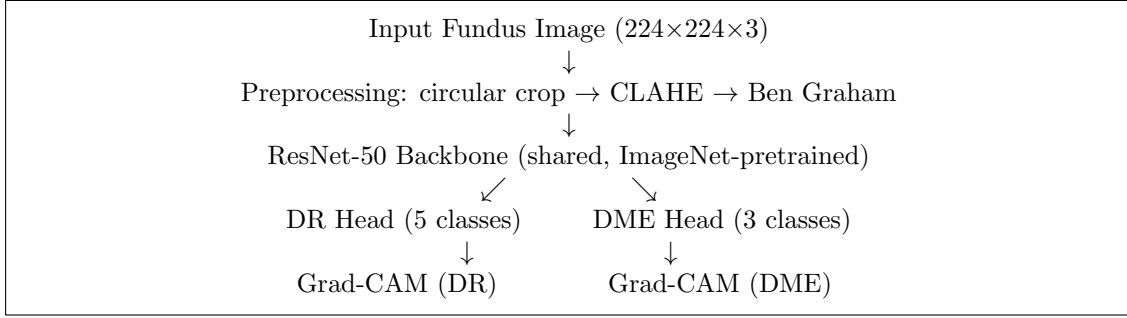


Figure 1: Shared-backbone, dual-head architecture.

3.3 Training

The model was trained with 5-fold stratified cross-validation (stratified on DR grade), AdamW optimization ($\text{lr} = 1 \times 10^{-4}$, weight decay $= 1 \times 10^{-5}$), and early stopping (patience 8) on the mean of the two tasks’ quadratic weighted kappa. Standard augmentations (flips, rotation, mild color jitter, Gaussian noise) were applied conservatively, since color carries diagnostic meaning in fundus images.

3.4 Explainability

Grad-CAM [2] was computed independently for each head by wrapping the model so gradients propagate only through the relevant head’s output logits. This isolates each task’s attention map and allows a direct test of whether the shared backbone produces task-differentiated features.

4 Results

Task	Quadratic Weighted Kappa	Macro ROC-AUC (OvR)
DR Grade (5-class)	0.657	0.743
DME Risk (3-class)	0.636	0.819
Combined (5-fold CV mean)	0.795 ± 0.038	

Table 1: Test-set and cross-validation performance.

Quadratic weighted kappa is the standard metric for ordinal medical grading tasks because it penalizes predictions proportionally to their distance from the true grade, rather than treating all misclassifications equally as raw accuracy does. These kappa values are consistent with published IDrID baselines for single-task DR grading, despite this model solving the additional, harder joint-task problem.

4.1 Finding 1: Severity Under-Grading

Per-class analysis of the confusion matrix revealed that severe cases (DR grade 4) are disproportionately misclassified as moderate (DR grade 2) rather than being confused with adjacent grades, as would be expected under a well-calibrated ordinal model. This is the more clinically dangerous failure direction: a model that under-grades

severity risks delaying urgent referral, whereas over-grading would at worst cause an unnecessary follow-up. Grad-CAM inspection of these misclassified cases showed attention concentrated on a subset of central lesions, with reduced sensitivity to peripheral hemorrhages that materially affect clinical severity grading.

4.2 Finding 2: Convergent, Not Divergent, Attention

The initial hypothesis was that DR attention (expected to be scattered across the retina, following microaneurysms and hemorrhages) and DME attention (expected to localize to the macular region specifically) would diverge, providing evidence of task-differentiated feature learning. Across multiple test cases, including both a disease-free image and a severe co-occurring DR/DME case, the two heads’ Grad-CAM maps were instead substantially overlapping, both concentrating on the same dominant lesion or optic-disc region. This suggests the shared backbone may be learning a single generalized “abnormality” representation rather than anatomically task-specific features, a plausible consequence of the small dataset size limiting the model’s ability to specialize per task. This is reported as a genuine, unexpected finding rather than the originally hypothesized result.

5 Limitations

- IDRiD’s Disease Grading subset is small (413 training images), which constrains both raw performance and the model’s capacity to learn task-differentiated representations.
- DR grade 1 and DME grade 1 are underrepresented in both training and test sets, and the model’s precision/recall on these minority classes is correspondingly weak.
- Grad-CAM indicates *where* the model attended, not *why*; it is a debugging and trust-building aid, not a validated clinical decision-support signal.
- Cross-entropy loss does not explicitly encode the ordinal structure of the grading scales; an ordinal-aware loss (e.g. CORAL) may further reduce the severity under-grading behavior identified above.

6 Conclusion

This project demonstrates that a shared-backbone, dual-head architecture can jointly grade DR and DME with performance comparable to single-task baselines, while surfacing two findings that would not be visible from accuracy metrics alone: a clinically consequential severity under-grading pattern, and an unexpected lack of task-specific attention divergence. Both findings motivate concrete next steps — ordinal-aware loss functions and task-specific fine-tuning of later backbone layers — and illustrate why explainability tooling belongs in the training and evaluation loop of clinical ML systems, not just as a post-hoc visualization layer.

References

- [1] Graham, B. (2015). Kaggle Diabetic Retinopathy Detection competition report.
- [2] Selvaraju, R. R. et al. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. *ICCV*.
- [3] Porwal, P. et al. (2018). Indian Diabetic Retinopathy Image Dataset (IDRiD). *IEEE DataPort*.